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To:
The Editors-in-Chief and Editorial Board
Data in Brief, Elsevier BV

Subject: Submission of Original Data Article — *IC4SD-TrafficSnap*

Article Title:
IC4SD-TrafficSnap: A City-Scale Multi-Camera Image Time-Series and Geospatial Road Network Dataset for Heterogeneous Urban Traffic

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Permanent Digital Object Identifiers:
- Target Dataset (Release v2.0): Zenodo DOI 10.5281/zenodo.22929940
- Companion Pilot Benchmark (Release v1.0): Zenodo DOI 10.5281/zenodo.22089673
- Companion Research Preprint: arXiv:2602.04159

Dear Editors-in-Chief and Editorial Board,

We are pleased to submit our original data article entitled **"IC4SD-TrafficSnap: A City-Scale Multi-Camera Image Time-Series and Geospatial Road Network Dataset for Heterogeneous Urban Traffic"** for consideration in *Data in Brief*.

Existing intelligent transportation benchmarks predominantly focus on either 1D inductive loop sensor time-series (e.g., METR-LA, PEMS-BAY) or isolated microscopic surveillance clips (e.g., UA-DETRAC, CityFlow) collected in car-dominant Western metropolitan areas. **IC4SD-TrafficSnap** addresses a critical global benchmark void by coupling a massive continuous visual surveillance snapshot archive with an OpenStreetMap-derived directed spatial road network in Ho Chi Minh City, Vietnam, where motorcycles account for 70–80% of modal vehicular volume:

- **714,123 continuous JPEG snapshots** (512×288 px, 44.38 GiB / 47.66 GB) recorded across **608 active stations** over **93.6 continuous hours** (October 2–6, 2026).
- **2,450 directed arterial corridor links** within a 6.0 km OSRM driving distance horizon, capturing real-world physical directionality (1,070 unidirectional links and 238 distance-asymmetric pairs).
- **Production PyTorch loaders, samplers, and dual-transition directed diffusion operators** (P_f, P_b, and scaled Chebyshev Laplacian $\tilde{L}$) ready for immediate self-supervised ViT pre-training and spatial graph learning.

Rationale for Three-Tier Licensing and the CC BY-NC 4.0 License

We understand that *Data in Brief* typically favors unrestricted open data reuse. To maximize open scientific reproducibility while strictly upholding public civic ethics and personal data protection regulations, the repository adopts an intentional **three-tier licensing framework**:

1. **Visual Snapshot Corpus (CC BY-NC 4.0):**
   - *Public Civic Purpose:* Snapshots were recorded from the open public traffic camera streams broadcast by the Ho Chi Minh City Department of Transportation (https://giaothong.hochiminhcity.gov.vn). Under Vietnamese Decree No. 47/2020/ND-CP on open government data sharing, this service was established specifically for civic travel information, congestion monitoring, and flood tracking.
   - *Ethical Non-Commercial Stewardship:* Because these camera streams are taxpayer-funded municipal assets deployed for community safety, applying the NonCommercial (NC) clause is a mandatory ethical obligation to prevent commercial corporations from appropriating and commercializing public surveillance feeds for private financial profit.
   - *Unrestricted Academic Openness:* The CC BY-NC 4.0 license guarantees 100% free, unrestricted access for all academic researchers, universities, non-profit institutions, and students globally to download, extract visual representations, benchmark machine learning models, and publish scientific findings.

- *Ethical Safety Layer for Privacy:* In addition to statutory sub-Nyquist anonymity ($p \leq 0.30\%$), the NC clause provides an extra ethical safeguard preventing data ingestion into commercial surveillance or face-matching products.

2. **Geospatial Road Graph and Distance Tensors (ODbL 1.0):** Fully complies with upstream OpenStreetMap licensing, ensuring open database reuse for all graph learning applications.
3. **Software Utilities and PyTorch Loaders (MIT License):** Permissive open-source license enabling seamless integration into deep learning toolchains.

Visual Privacy-by-Design and Empirical Multi-Tier Audit

The visual corpus captures macroscopic urban traffic dynamics and guarantees anonymity by design:

- **Sub-Nyquist Optical Limits:** High mounting elevations ($H \geq 6.0$ m) and native 512×288 resolution restrict license plate character stroke heights to at most 2.5–2.8 pixels (far below ALPR readability of ≥ 16 px).
- **Biometric Occlusion:** Statutory motorcycle helmets (Decree 100/2019/ND-CP) and protective fabric sun/dust masks ($> 85\%$) physically occlude facial biometrics.
- **Statistical Rule of Three ($p \leq 0.30\%$):** Across $3,000$ audited snapshots (including $1,000$ worst-case daylight stress frames evaluated by three independent raters with 100% unanimous agreement), zero readable license plates or facial features were detected. By the Hanley and Lippman-Hand Rule of Three, the 95% upper confidence bound on residual privacy risk is bounded at $p \leq 0.30\%$.
- **Three-Tier Revocation Protocol:** An enforceable mechanism comprising an official public registry (`REVOKED_STATIONS.csv`), an automated local purge script (`code/redact_revoked_stations.py`), and Zenodo minor version deprecation.

Dataset Lineage and Companion Pilot Benchmark

The dataset evolved from an initial pilot benchmark (*IC4SD-Traffic-HCM*, v1.0, Zenodo DOI: 10.5281/zenodo.22089673), which provided 5,012 annotated frames and 86-day flow series. **IC4SD-TrafficSnap** represents a major city-scale expansion ($140\times$ visual corpus scaling to 714,123 frames and a 2,450-edge directed road graph). Both the pilot Zenodo record and the companion preprint (arXiv:2602.04159) are fully documented in the manuscript.

We confirm that this manuscript adheres strictly to the *Data in Brief* guidelines and represents original data documentation. Thank you for your time and consideration.

Sincerely,

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